

Fruit fly assay report

Session: /home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_132535

Run dir: /home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_132535

Raw video:

/home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_132535/raw_video.mp4

Background:

/home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_132535/background_transformed_snapshot.png

Calibration:

/home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_132535/calibration_snapshot.json

Duration (s): 9.8

Frames processed: 50

Tracking mode: stitch

Mask blobs as flies: True

Assay tubes: 4

Track fragments detected: 17

Unique threshold crossings: 3

0 s to 9.8 s comparison window

0 s to 9.8 s x-displacement summary
Columns 1-6 of 9 | Rows 1-4 of 4

assay_tube_index	flies_compared	mean_abs_dx_px	median_abs_dx_px	mean_signed_dx_px	comparison_target_s
1	8	9.75	4.9	6.389	9.8
2	1	9.79	9.79	9.79	9.8
3	1	0.103	0.103	0.103	9.8
4	3	7.478	1.488	-5.788	9.8

0 s to 9.8 s x-displacement summary
Columns 7-9 of 9 | Rows 1-4 of 4

mean_abs_dx_mm	median_abs_dx_mm	mean_signed_dx_mm
0.942	0.473	0.617
1.036	1.036	1.036
0.011	0.011	0.011
0.817	0.163	-0.633

Threshold crossing summary
Columns 1-5 of 5 | Rows 1-4 of 4

assay_tube_index	number_of_flies_detected	number_of_unique_threshold_crossings	fraction_crossing_by_10s	first_crossing_time_s
1	10	3	0.3	1.4
2	1	0	0.0	nan
3	3	0	0.0	nan
4	3	0	0.0	nan

Threshold crossing events
Columns 1-6 of 8 | Rows 1-3 of 3

unique_event_id	assay_tube_index	display_id	crossing_time_s	crossing_x_px	crossing_y_px
1	1	7	1.4	146.833	232.583
2	1	2	3.8	148.217	236.101
3	1	3	5.4	148.375	264.562

Threshold crossing events
Columns 7-8 of 8 | Rows 1-3 of 3

threshold_distance_px	threshold_distance_mm
501.0	66.912
501.0	66.912
501.0	66.912

Velocity summary
Columns 1-6 of 6 | Rows 1-4 of 4

assay_tube_index	number_of_flies_detected	mean_speed_mm_s	max_speed_mm_s	mean_vertical_velocity_mm_s	max_vertical_velocity_mm_s
1	10	7.918	79.267	-0.27	73.045
2	1	0.585	2.812	-0.337	0.499
3	3	2.505	7.697	0.009	2.79
4	3	0.797	9.434	-0.694	0.077